

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T40 · Angles in Polygons and Parallel Lines

With Answers

Questions with full worked solutions

■ 10 pages

FAMILY MAP

Angles in Polygons Parallel Lines

F01 Use quadrilateral properties

2 Q | 9 marks

F02 Use triangle angle properties

1 Q | 4 marks

F03 Use basic angle and line facts

1 Q | 5 marks

Question 14

4MA1-2H Q3

ORIGINAL QUESTION

4 marks

- 3 $ABCD$ is a trapezium.

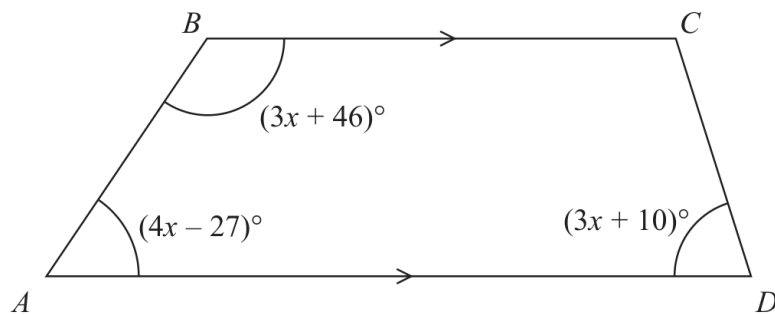


Diagram **NOT**
accurately drawn

BC is parallel to AD

Find the size of the largest angle inside the trapezium.

Worked Solution 14

Largest angle in a trapezium

MODULAR UNIT 1

4 marks

Use parallel-line angles

$$(4x - 27) + (3x + 46) = 180 \Rightarrow 7x + 19 = 180 \Rightarrow x = 23.$$

Evaluate the angles

$$3x + 46 = 115, \quad 3x + 10 = 79, \quad 4x - 27 = 65; \text{ the remaining angle is } 101.$$

Final answer

$$115^\circ$$

- 4 The diagram shows a triangle.

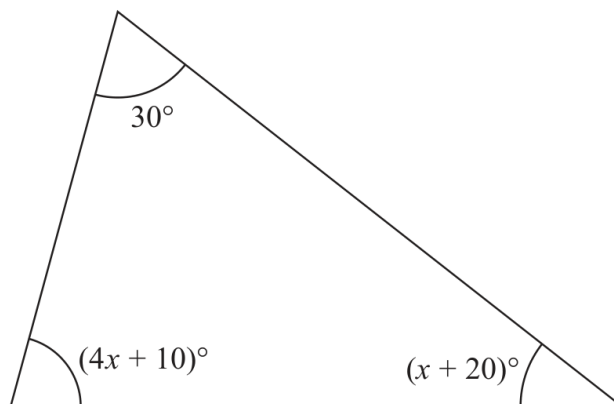


Diagram **NOT**
accurately drawn

Work out the value of x .

$x = \dots\dots\dots$

Worked solution

January 2020 | Question 4 | 4 marks

Family: Use triangle angle properties • Variant: Solve an algebraic triangle-angle chain with reasons

Method / working

1. Triangle angles sum to 180° .
2. $30 + (4x + 10) + (x + 20) = 180$.
3. $5x + 60 = 180$, so $5x = 120$.
4. $x = 24$.

Final answer: $x = 24$

5

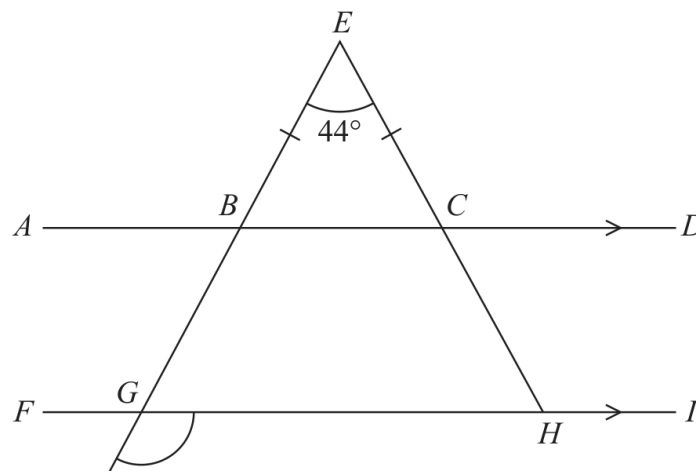


Diagram **NOT**
accurately drawn

$ABCD$ and $FGHI$ are parallel straight lines.
 $EBGJ$ and ECH are straight lines.

$BE = CE$
Angle $BEC = 44^\circ$

Work out the size of angle JGH .
Give a reason for each stage of your working.

o

Worked solution

June 2021 | Question 5 | 5 marks

Family: Use basic angle and line facts • Variant: Complete a multi-step line-angle reason chain

Method / working

1. $BE=CE$, so base angles of triangle BEC are $(180-44)/2=68^\circ$.
2. Use the straight line at B: the adjacent angle is $180-68=112^\circ$.
3. Transfer the angle through the parallel lines using corresponding/alternate angles.
4. Use the straight-line relation at G to reach angle JGH= 112° .
5. State the isosceles, straight-line, parallel and angle-sum reasons.

Final answer: 112°

3

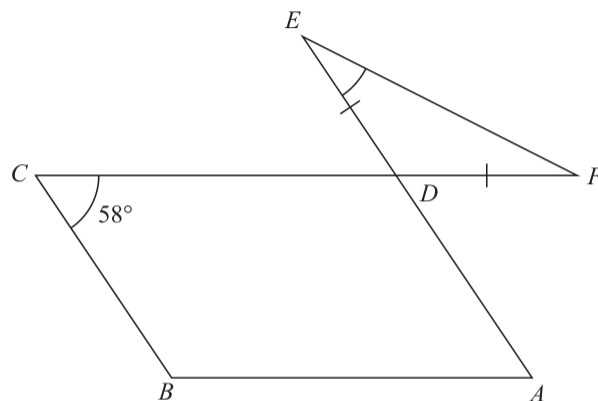


Diagram **NOT**
accurately drawn

The diagram shows a parallelogram $ABCD$ and an isosceles triangle DEF in which $DE = DF$

CDF and ADE are straight lines.

Angle $BCD = 58^\circ$

Work out the size of angle DEF .

Give a reason for each stage of your working.

(Total for Question 3 is 5 marks)

Answer reference | May-Nov 2020 | Question 3

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

3	$ADC = 180 - 58 (= 122)$ or $EDF = 122$ or $CDE = 58$ or $ADF = 58$			M1 may be seen marked on the diagram
	e.g. $DEF = 58 \div 2$ or $DEF = (180 - 122) \div 2$			M1 complete method to find angle DEF
		29		A1
			5	B2 dep on M2 for fully correct reasons for their method (B1 dep on M1 for one correct reason stated and used) e.g. <u>Allied angles</u> , <u>co-interior angles</u> , <u>Alternate angles</u> , <u>Corresponding angles</u> , <u>Vertically opposite angles</u> are equal (or Vertically <u>opposite angles</u> are equal), <u>Angles</u> on a straight <u>line</u> add up to 180° (or angles on a straight <u>line</u> add to 180°), Sum of <u>two angles</u> in a triangle are equal to <u>opposite exterior angle</u> , <u>Angles</u> in a <u>triangle</u> add up to 180° (or Angles in a <u>triangle</u> add up to 180°), Base angles in an <u>isosceles triangle</u> <u>Angles</u> in a <u>quadrilateral</u> add up to 360. (accept "4- sided shape" or parallelogram) <u>Opposite angles</u> of a <u>parallelogram</u> are equal
Total 5 marks				

Worked solution

May-Nov 2020 | Question 3 | 5 marks

Family: Use quadrilateral properties • Variant: Give standard reasons in a quadrilateral angle chain

Official final mark-scheme pages are embedded immediately after this question.